Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer pursuing an MSCS at USC with 2+ years building reliable systems in **C++/Python-adjacent environments**, with hands-on experience in **Linux**, testing, CI/CD, simulation, and production-quality debugging for complex software workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Operating Systems, Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Embedded Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **platform services and validation workflows** for 500+ users, reducing request latency from **1.2s to 350ms** while improving issue diagnosis, operational visibility, and responsiveness across production software systems.
- Implemented **automated validation and ETL checks** for 5,000+ monthly jobs, reducing downstream failures while improving monitoring, release confidence, and production support readiness across recurring software workflows.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **large-scale simulation pipelines** across 100,000+ trajectory records, enabling reproducible algorithm evaluation, faster diagnostics, and high-confidence benchmarking while improving signal quality for debugging model-driven control workflows.
- Architected a **spatial-temporal encoder** integrating map geometry with agent context, improving closed-loop planning stability by **15%** while strengthening evaluation rigor for long-horizon autonomy experiments.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API and ETL microservices** using AWS Glue and S3, improving response time by **30%** while supporting distributed integrations, cross-platform validation, and stable delivery across production workloads.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing CI reliability by **68%**, reducing regressions, and strengthening release quality for coordinated software components.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **MERN and WebSocket modules** for 2M+ synchronized records across eight workflows, improving throughput consistency, reducing stale-session behavior, and strengthening responsiveness for real-time user-facing applications.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%**, improving deployment consistency, and reducing environment drift that previously slowed validation and troubleshooting.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **backend features and APIs** handling 10,000+ daily requests, translating product requirements into maintainable services while improving supportability, implementation ownership, and production debugging workflows.
- Built **AI-assisted grading workflows** with multimodal models and OCR, reducing feedback turnaround by **60%** while supporting production operations through concurrent execution, CI/CD-backed releases, and cost-aware debugging.

Skills

Languages: C++, Python, Java, JavaScript/TypeScript, SQL, Bash
Embedded & Systems: Linux, multithreaded environments, CMake/Make familiarity, low-level debugging, system integration
Protocols & Interfaces: TCP/IP, JSON, YAML, hardware-software integration concepts
Backend & Tooling: Flask, FastAPI, REST APIs, Jenkins, GitHub, CI/CD, automated testing
Cloud & Infrastructure: AWS, Azure, Docker, Kubernetes, cross-platform validation workflows
Engineering: Agile/Scrum, technical documentation, code reviews, reliability-focused development

Projects

CPU-scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, priority, and MLQ** algorithms in a **C++**-backed simulator, pairing interactive controls with Gantt visualization across 1K+ process combinations.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, improving reliability, reducing deployment variance, and streamlining troubleshooting across production workflows.

HMIS Live | *MERN, TypeScript, Twilio, Appwrite*

- Built a **workflow-heavy operations platform** with scheduling, notifications, and dashboard experiences, reducing administrative workload by **30%** through secure automation and structured system behavior.